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Lab 03.2: Analysis of consumption for Plug-in Hybrid Electric Vehicles (PHEVs)

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Abstract

This report details the analysis of the performance and consumption of a Plug-in Hybrid Electric Vehicle (PHEV) across the WLTP (Worldwide Harmonized Light-Duty Vehicles Test Procedure) driving cycle. The study considers various operating strategies for the two power sources: the Internal Combustion Engine (ICE) and the Electric Motor (EM).

The primary objective of this work is to evaluate the vehicle's overall efficiency and the management of the Battery State of Charge (SOC) during different phases of the cycle.

The analysis distinguished the powertrain behavior across two main segments:

1. **Electric Motor (EM) Drive (Urban Segment):** the EM demonstrated high efficiency during traction but suffered from significantly low efficiency during the regenerative braking phase
2. **Internal Combustion Engine (ICE) Drive (Extra-Urban Segment):** two distinct strategies were examined:
    - **Force Balancing:** the ICE provided only the necessary power for traction, resulting in a near-constant battery SOC. This approach highlighted the inherent low efficiency of the ICE when operating outside its optimal range
    - **Constant Power (Surplus):** the ICE operated at a constant, efficient power level, generating an energy surplus that was continuously diverted to recharge the battery. This led to a linear increase in the SOC over time

The results quantify the effectiveness of the hybrid architecture in exploiting the advantages of each motor to optimize consumption throughout the WLTP homologation cycle, demonstrating how power flow management directly influences the vehicle's efficiency and range.

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1 Introduction

The aim of this lab is to analyze the performance of a plug-in hybrid electric vehicle in different working conditions. The powertrain layout of the vehicle is represented in Figure 1

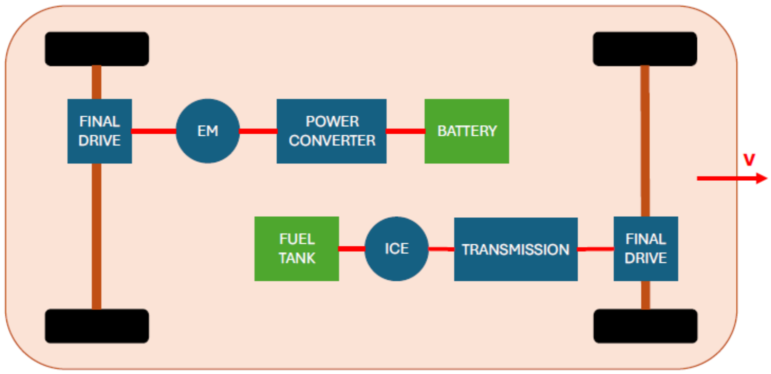

Figure 1

- The electric motor powers the rear axle
- The Internal Combustion Engine powers the front axle

The main idea behind the usage of a hybrid vehicle is to exploit the main advantages both ICE and of EM:

- ICE can provide higher power but can only work within a limited rpm range and it has low efficiency in general
- EM can provide less power but it can work even with low rpm and has a higher efficiency

$\Longrightarrow$ goal: use the EM at low rpm and the ICE in the rpm range where it is more efficient

The vehicle considered can operate in different modes:

- ICE drive mode → traction torque is delivered by the ICE only
- Electric drive mode → traction torque delivered by EM only
- Hybrid drive mode → traction torque delivered by both ICE and EM (ICE operates in the region of maximum efficiency)
- Energy recovery mode → the EM works as a generator and provides the power to charge the battery

To test the performance of the car, a vehicle speed cycle is imposed (Figure 2). The velocity profile is defined by the *Worldwide Harmonized Light-Duty Vehicles Test Procedure (WLTP)*; it is a global, harmonized standard for assessing the performances of light-duty road vehicles in terms of fuel consumption, pollutant levels and CO2 emissions. Since 2019 all light duty vehicles are tested according to this standard.

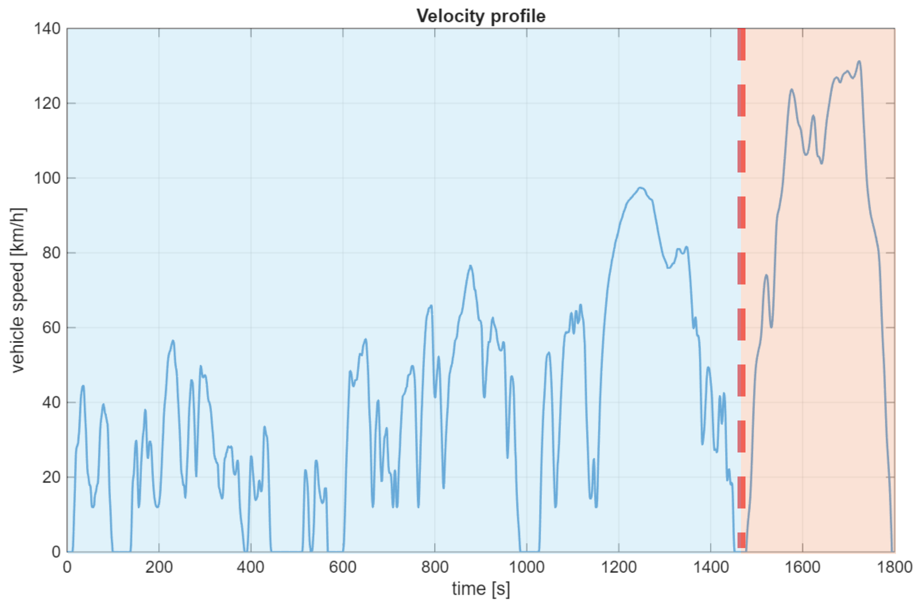

Figure 2

This cycle has been defined based on the ratio between the power and the unladen mass of the vehicle; in the considered case, the vehicle belongs to "class 3". The velocity profile is supposed to simulate the behavior of the vehicle in different scenarios. It can be divided into two portions:

- **The first region (EM drive)** is characterized by low values of speed but frequent changes (to simulate driving in urban environment).
    - In this region the vehicle operates in EM mode (only the electric motor is considered)
    - **Case 1.1**:
        - If the required power at the shaft is > 0 → battery discharges and provides power to the wheels
        - If the required power at the shaft is < 0 → regenerative drive mode is engaged and the battery charges
    - **The performance of the vehicle is evaluated in terms of electric energy consumption**
- **The second region (ICE drive)** is characterized by high values of speed and few changes (to simulate driving on extra urban roads).
    - Two different cases are identified based on the coupling between the EM and the ICE

- **Case 2.1:** ICE provides just enough power to balance the power of the resistance forces; the battery is recharged only during braking
  - If the required power at the shaft is > 0 → ICE provides power
  - If the required power at the shaft is < 0 → battery recharges
- **Case 2.2:** ICE provides constant power at maximum efficiency
  - If the ICE power is higher than the power of the resistance forces (the required power) → ICE recharges the battery
  - If the ICE power is lower than the power of the resistance forces (the required power) → EM also provides power together with ICE
  - If the required power is negative → battery recharges
- **In both cases, the performance of the vehicle is evaluated in terms of electric energy consumption, fuel energy consumption and battery state of charge (%SOC)**

2 EM drive

The first region of Figure 2 is considered. The goal is not to compute the average performance in the whole interval; to get more precise results, the time history has been divided into intervals of 1s each and the procedure described below has been repeated for each interval:

- Computation of the total resistance forces R_{tot} given by the sum of the rolling resistance, the aerodynamic resistance and the inertia force
- Computation of the required power at the wheels: $P_{diss} = R_{tot} * v$
- Computation of the required power at the shaft by using the efficiency of the transmission (the efficiency of the transmission has been assumed to be 0.97 based): $P_{shaft} = P_{diss}/\eta_t$
- Computation of the required electrical power:
  - $P_{shaft} > 0 \rightarrow P_{el} = \frac{P_{shaft}}{\eta_{EM}} > 0$ (battery discharges)
  - $P_{shaft} < 0 \rightarrow P_{el} = \frac{P_{shaft}}{\eta_{REG}} < 0$ (battery recharges)
- NB: the regeneration efficiency has been assumed to be $\eta_{REG} = 0{,}3$ while the electric motor efficiency needs to be estimated from the diagram in Figure 3 as a function of the motor speed and the torque that the EM needs to provide ($\eta_{EM} = \eta_{EM}(RPM, T)$):

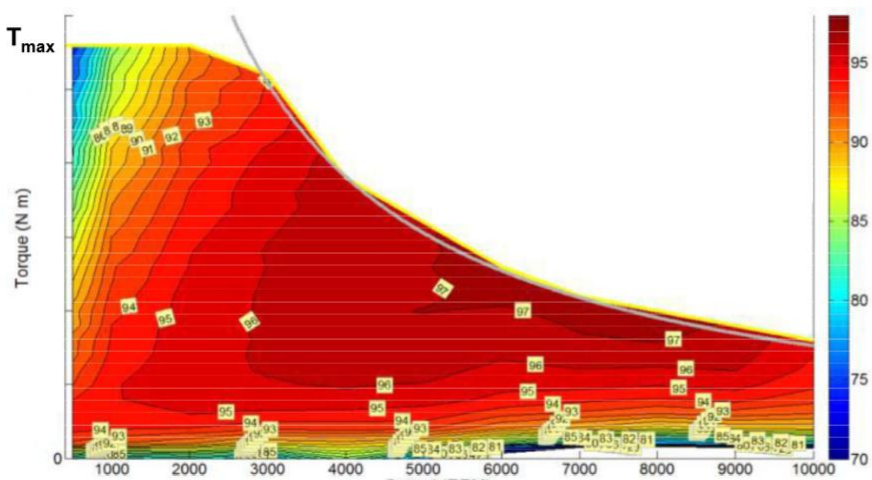

Figure 3

2.1 Case 1.1

After having performed the computations described previously for all the time intervals, the results have been collected in vectors and plotted. Figure 4 displays the power of the resistance forces as a function of speed:

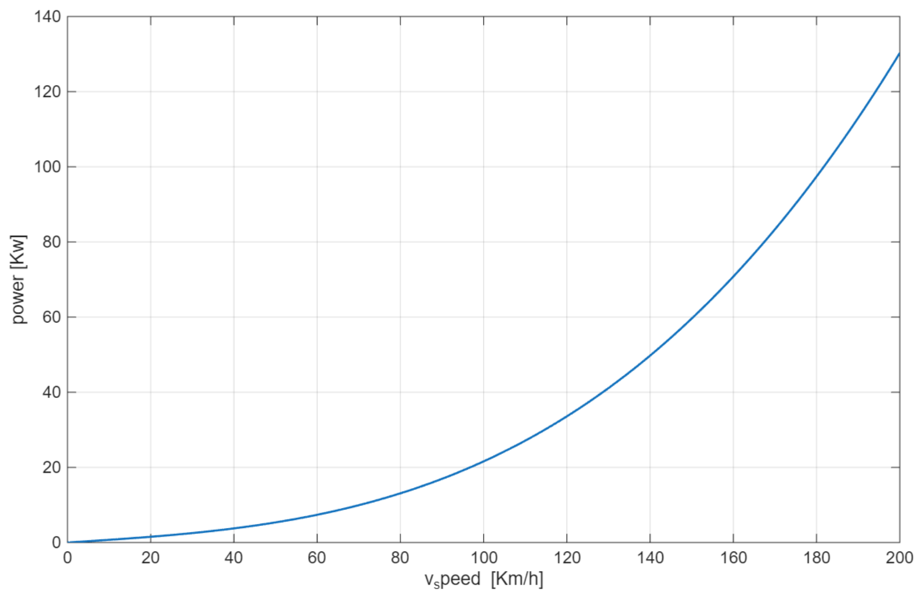

Figure 4

Figure 5 displays P_{el} (in green) and P_{shaft} (in blue). Some observations can be made by looking at a specific part of the cycle that is displayed just below:

- When the requested power is positive, the green diagram lies just above the blue one, this indicates that the EM is operating in a high efficiency condition (η_{EM} is close to 1
- When the requested power is negative, the distance between the diagrams is significant, highlighting the extremely low efficiency of regeneration compared to the electric motor efficiency in driving conditions

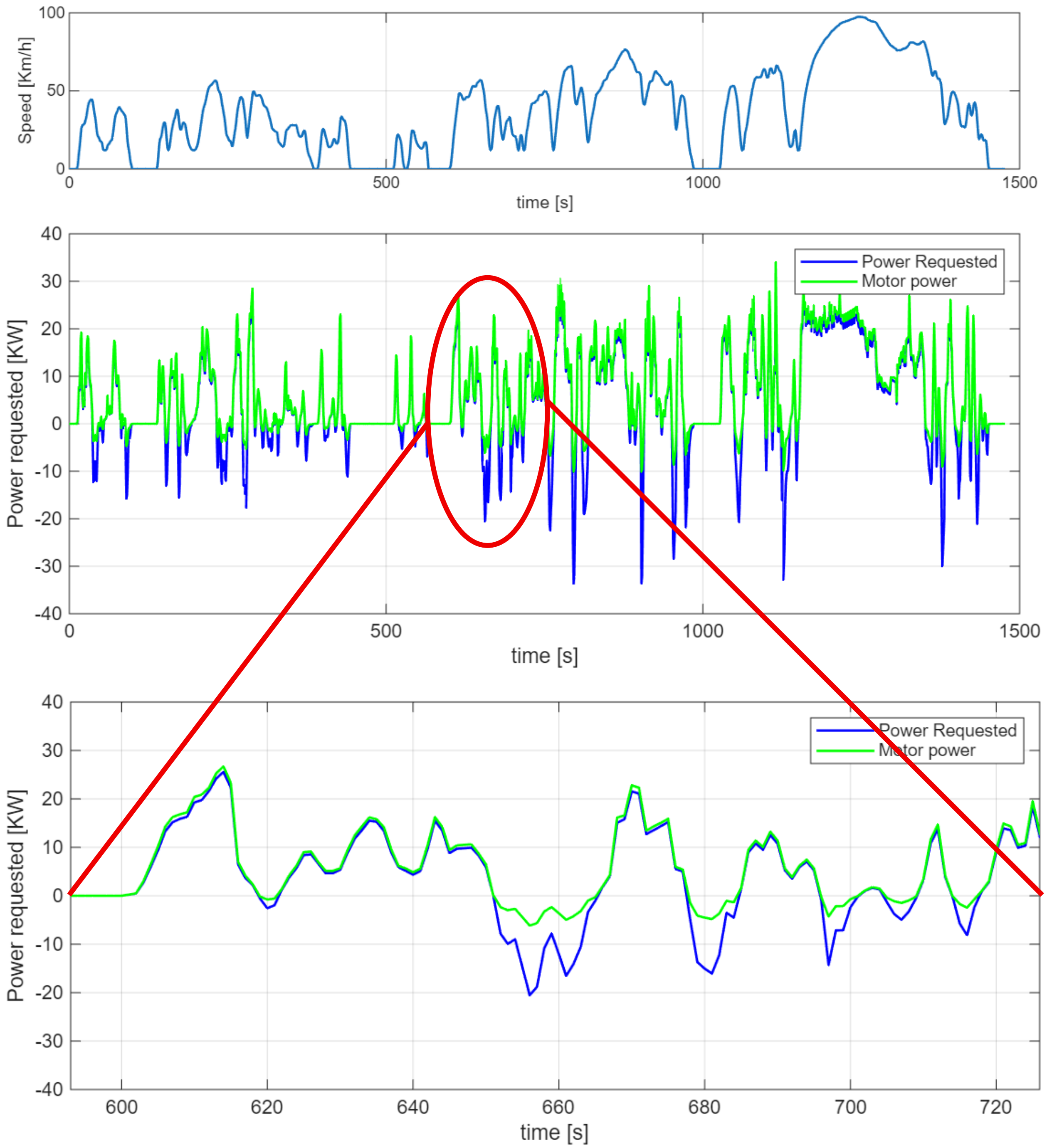
Speed [Km/h]
time [s]
Power requested [KW]
time [s]
Power Requested
Motor power
Power requested [KW]
time [s]
Power Requested
Motor power

Figure 5

3 ICE drive

The second region of Figure 2 is considered. As for EM computation of: total resistance forces R_{tot}, required power at the wheels P_{diss} and required power at the shaft P_{shaft}, was made in the same way.

Differently from quantities listed above:

- Computation of required power:
  - **Case 2.1**
    - $P_{shaft} > 0 \rightarrow P_{fuel} = \frac{P_{shaft}}{\eta_{ICE}}$ (ICE provides power)
    - $P_{shaft} < 0 \rightarrow P_{el} = P_{shaft} * \eta_{REG} < 0$ (battery recharges)
  - **Case 2.2**
    - $P_{shaft} > 0$
      - $P_{engine} > P_{shaft}$
        - $\Delta_{power} = P_{engine} - P_{shaft} > 0$ (ICE provides power and recharges battery)
        - $P_{fuel} = \frac{P_{shaft}}{\eta_{ICE}}$ (fuel consumption)
      - $P_{engine} < P_{shaft}$
        - $\Delta_{power} = P_{engine} - P_{shaft} < 0$ (ICE provides power together with battery that discharges)
        - $P_{fuel} = \frac{P_{shaft}}{\eta_{ICE}}$ (fuel consumption)
    - $P_{shaft} < 0$
      - $P_{el} = P_{shaft} * \eta_{REG} < 0$ (battery recharges)
- Battery state of charge (SOC):
  - Energy is computed as: $E_{el} = \int P_{el}(t)dt$
  - Variation of the battery SOC is given by: $SOC(t)[\%] = \frac{C_{batt} - E_{el,kWh}(t)}{C_{batt}}$, where C_{batt} is the battery capacity and $E_{el,kWh}(t)$ is the electric energy in kWh.
- Fuel tank:

  The fuel tank was modeled as an energy reservoir. The total available energy is calculated as follows:

  $$\text{Tank} = \text{m}_{\text{fuel}} * H_i$$

  Where:
  - m_{fuel} is the initial mass of fuel and
  - H_i is the lower calorific value of the fuel.

To calculate the remaining fuel percentage, the simulation updates the status at every second.
The energy required by the ICE is subtracted from the total energy remaining in the tank.

3.1 Case 2.1

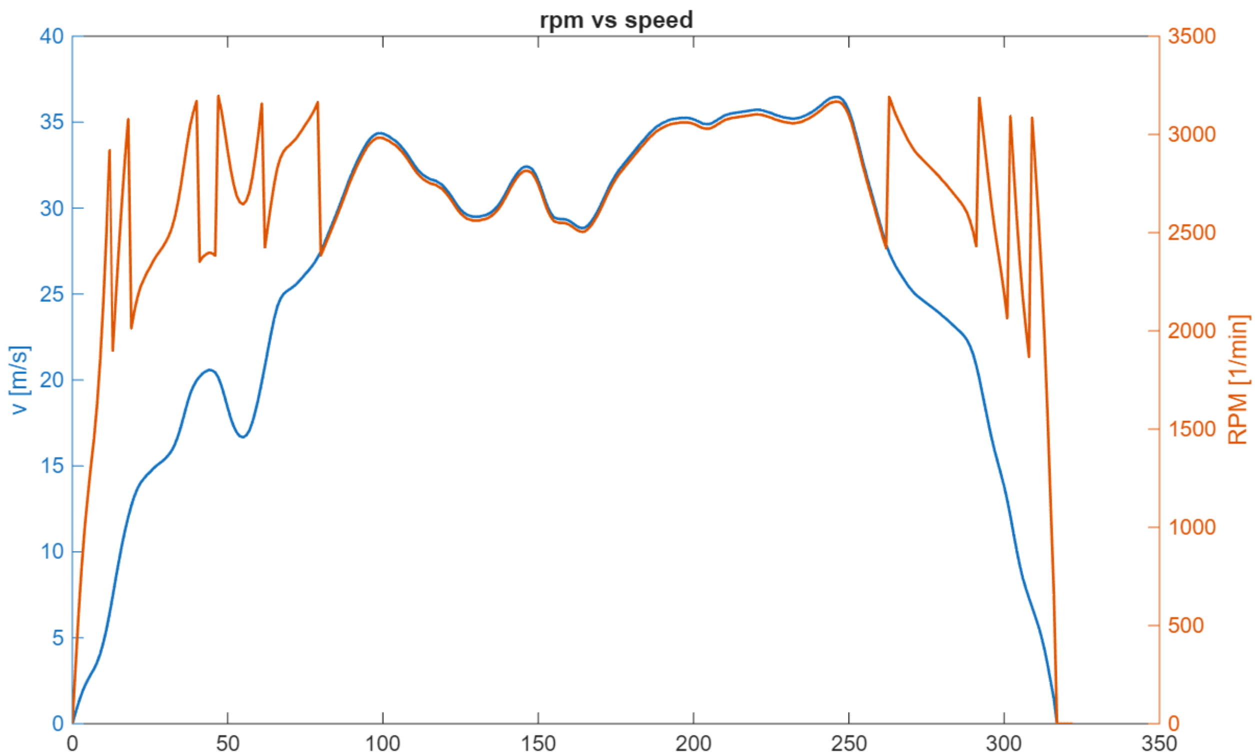

Figure 6

Figure 6 illustrates the relationship between vehicle speed and engine RPM, showing the corresponding gear shifts.

To achieve this, a gear change logic was implemented in our code, designed to select the optimal gear to maintain the engine RPM within an optimal range.

This optimal range, approximately 1800 to 3200 RPM, was determined based on the efficiency map of the ICE provided in the assignment.

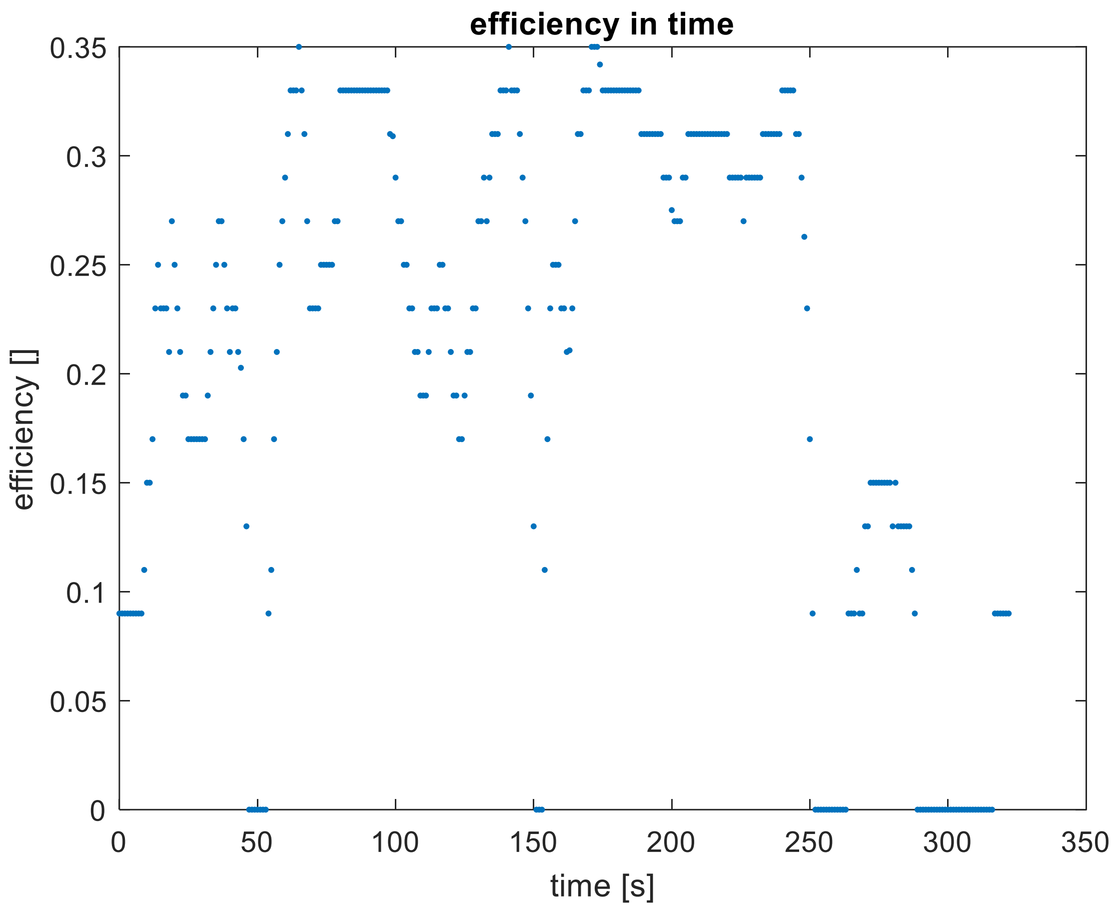

Figure 7

In **Errore. L'origine riferimento non è stata trovata.** it is possible to observe that even with a gearbox whose aim is to maximize operation at the highest efficiency this is not always possible.

Furthermore, it is evident that ICE has a significantly lower efficiency with respect to EM.

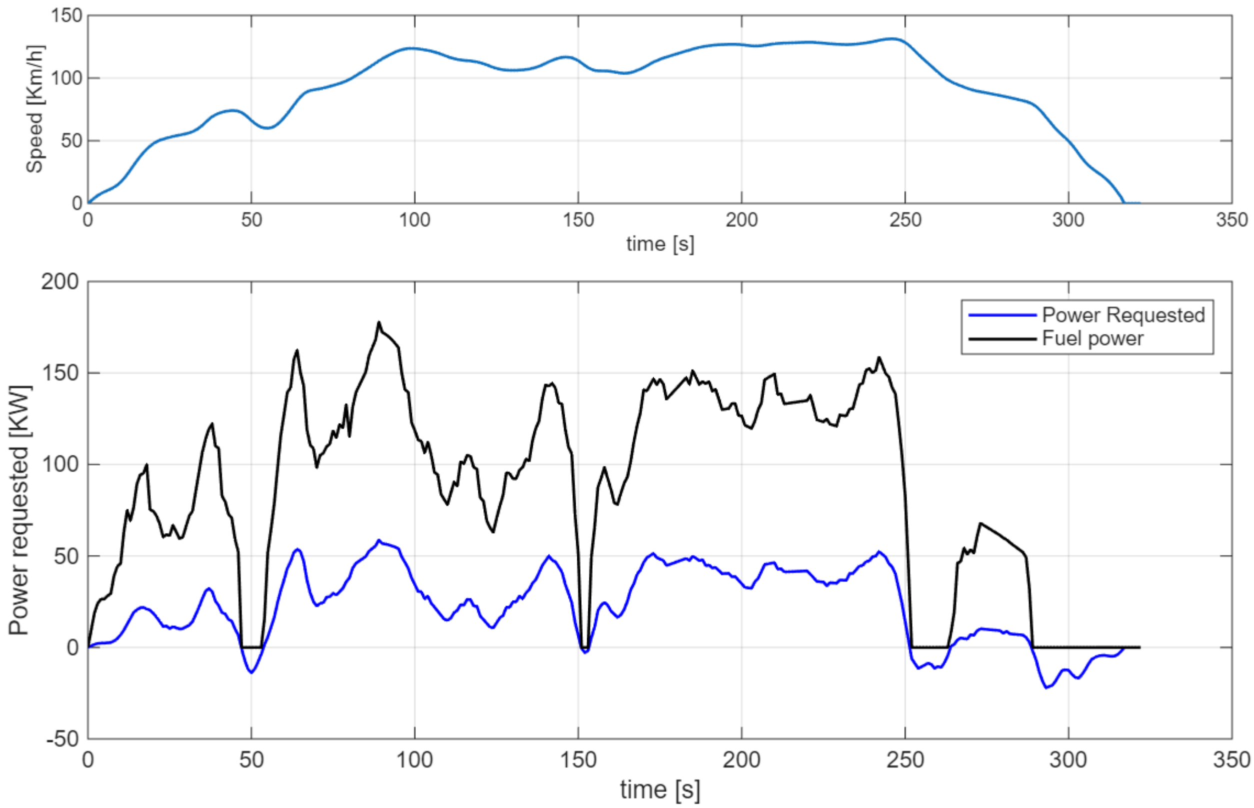

Figure 8

Figure 8 display the requested power and the fuel power needed to follow the speed profile above. The efficiency plot also confirms that the fuel power input must be higher than the requested output due to the low efficiency of the ICE.

To compute the power given by the fuel

$$P_{fuel} = \frac{P_{shaft}}{\eta_{ICE}}$$

The ICE efficiency under every condition is sourced directly from the performance map included as a reference in the assignment

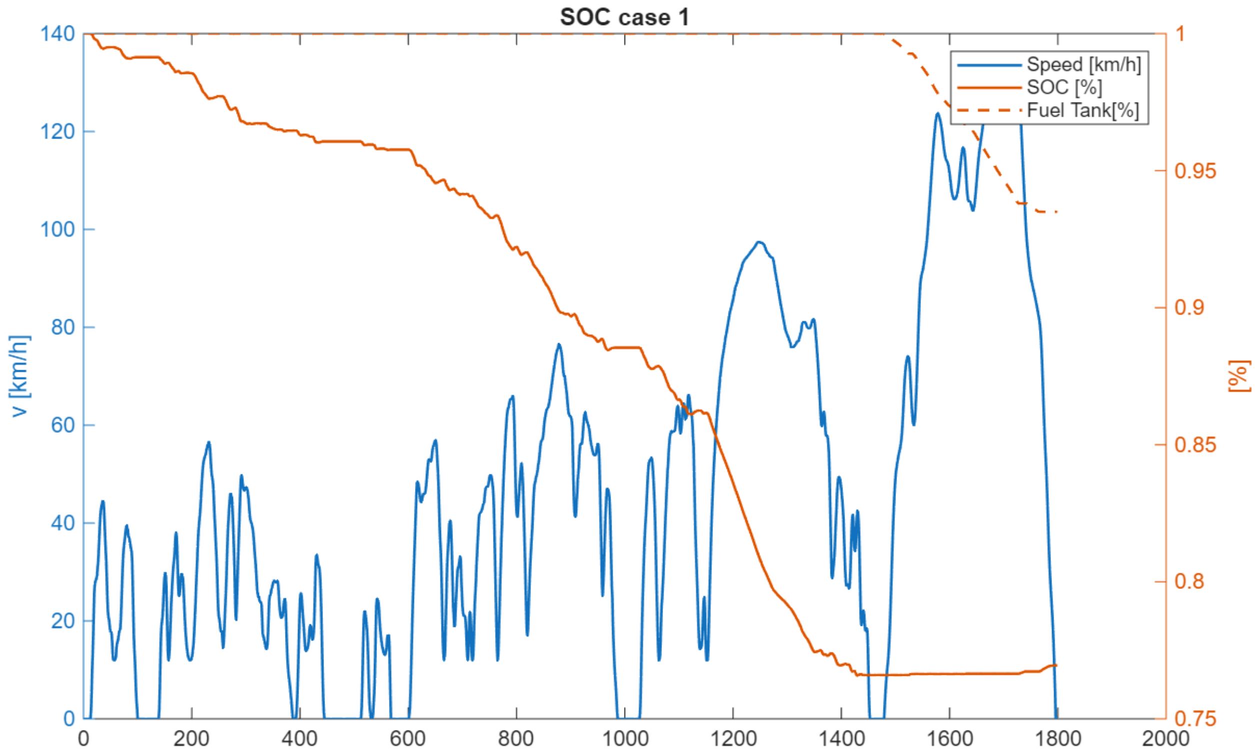

Figure 9

Figure 9 illustrates the SOC and the fuel tank percentage throughout the test, correlated with the required speed profile.

As expected, in the initial phase where only the EM is utilized, the SOC decreases constantly.

Conversely, starting from t=1500 s, the ICE begins operating. We observe that the fuel tank percentage decreases, while the SOC is maintained at a near-constant level,

This observation implies that the ICE provide just enough power to satisfy the immediate demand for the speed profile, with minimal or no surplus power used to significantly increase the battery SOC.

3.2 Case 2.2

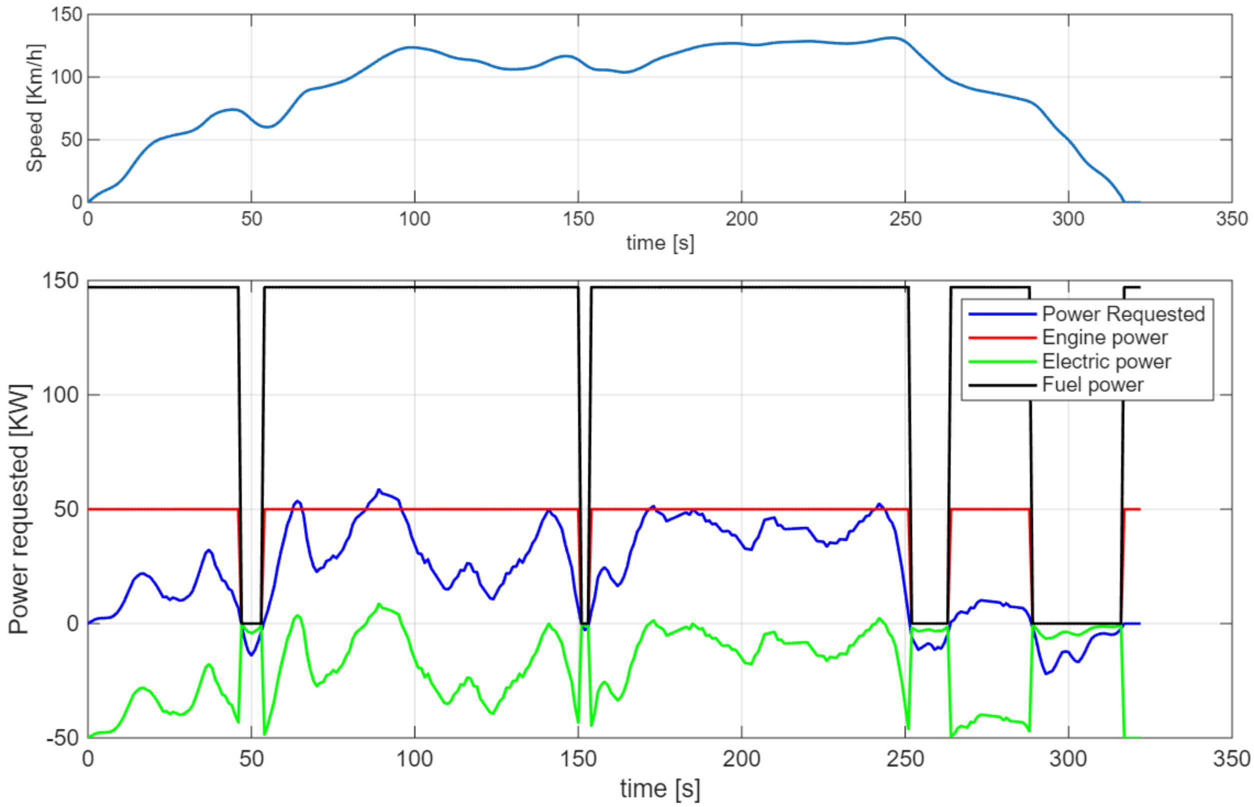

Figure 10

Figure 10 illustrates the power output trends for the ICE and the EM in a configuration where the ICE is set to operate at a constant RPM that ensures maximum efficiency.

It can be observed that, for the majority of the test cycle, the power generated by the ICE exceeds the power required to follow the vehicle's speed profile. This power surplus results in battery recharging, which is represented by a negative value for the electric power in the graph.

Only in rare instances, where the ICE power is insufficient to meet the traction demand, does the EM intervene to provide supplementary power to the system.

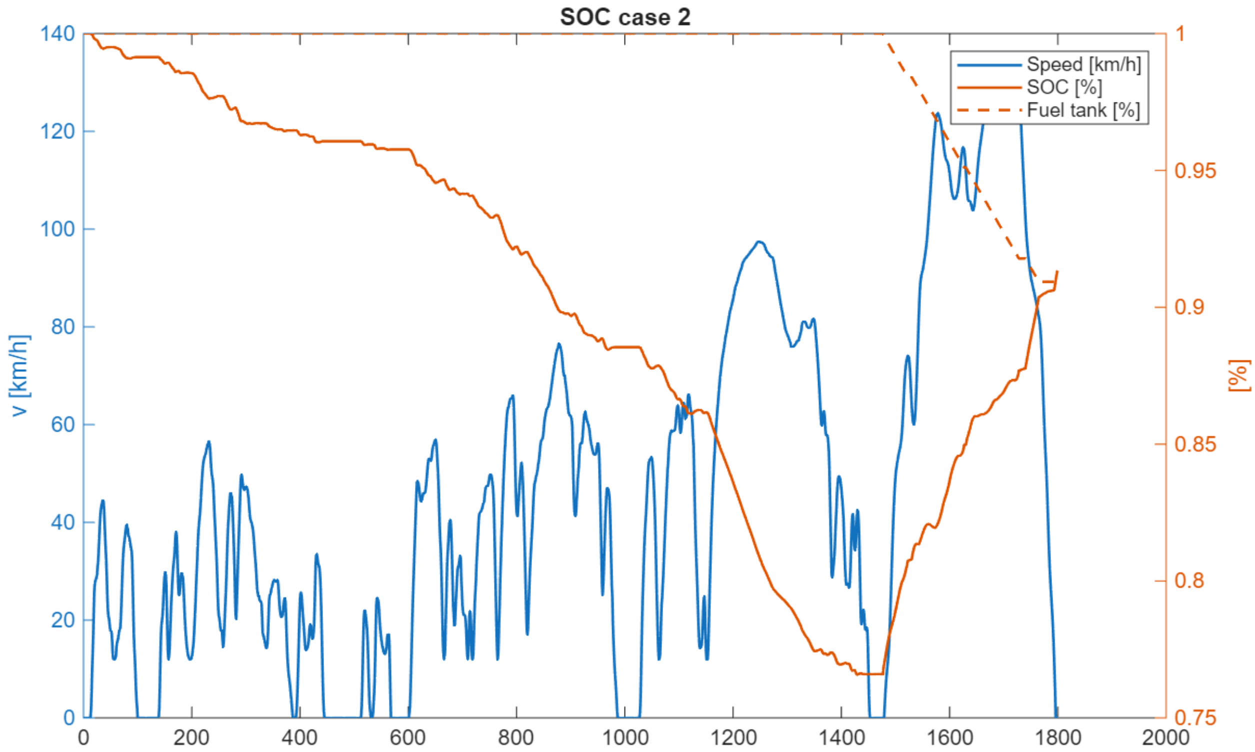

Figure 11

As we expected differently from Figure 9 when the ice start to work at constant speed, according to Figure 10, the SOC increase constantly during this part of the test and as a consequence the fuel tank percentage decrease linearly with respect to time

4 Conclusion

This lab analyses the performance of the Plug-in Hybrid Electric Vehicle (PHEV) across the WLTP speed cycle.

- **EM Drive (Urban Segment):** the Electric Motor (EM) shows high efficiency during propulsion but suffers from extremely low efficiency during regenerative braking
- **ICE Drive (Extra Urban Segment):**
  - **Case 2.1:** the Internal Combustion Engine (ICE) provides just enough power to balance the resistance forces while battery is recharged only during braking phases; this results in a near-constant battery State of Charge (SOC). Furthermore, the ICE's significantly low efficiency is highlighted by the large distance between the engine power and the fuel power
  - **Case 2.2 (Constant Efficiency):** the ICE operates at constant power, creating a power surplus that recharges the battery. This leads to a constant increase in SOC during the driving phase